ZEKA EMO

+62-85171614749 | zeka.emo30@gmail.com | www.linkedin.com/in/zekaemo | https://github.com/zekaemo | Portfolio

Computer Engineering graduate with experience spanning applied AI, data analysis, and software engineering. Built machine-learning workflows for object detection, Indonesian-language sentiment analysis, recommendation, and code retrieval, from data preparation and feature engineering through evaluation. At Grab, worked on Go and React monitoring tools, APIs, production debugging, and legacy migration; at PT Jamkrindo, developed PHP/MySQL backend workflows and database schema for a letter-tracking system. Also experienced with Python, PyTorch, TensorFlow, scikit-learn, and Optuna.

Work Experiences

Daegu Gyeongbuk Institute of Science and Technology (DGIST) – **Daegu, South Korea** Jun 2026 – Aug 2026
Research Intern
- Conducted research on graph-based bug localization, representing software repositories as graphs to capture structural relationships among code entities and identify potential bug locations.

Diponegoro University – **Semarang, Indonesia** Aug 2025 – Jun 2026
Research Assistant
- Reconstructed the YOLO architecture using C3Ghost modules to optimize lightweight small-object detection in SAR maritime imagery, including dataset preparation and model evaluation.
- Co-authored the full research manuscript for Q3-indexed journal submission, articulating technical insights and results in writing.

Grab – **Jakarta, Indonesia** Sept 2025 – Feb 2026
Full-Stack Engineer Intern
- Built and maintained internal data monitoring tools using Go and React, improving reliability through production debugging, legacy feature migration, and API development.

PT Jaminan Kredit Indonesia - **Bandung, Indonesia** Jul 2024 - Aug 2024
Back-End Developer Intern
- Developed and implemented a comprehensive letter management system, streamlining the tracking of incoming and outgoing correspondence for West Java Regional Office.

LuarSekolah - **Bandung, Indonesia** May 2023 - Jun 2023
Mobile Developer Intern
- Leveraged Flutter and Dart to design and develop a functional Android app prototype, incorporating 5 key widgets and 5 features.

Education Level

Universitas Diponegoro - **Semarang, Indonesia** Aug 2022 – Sept 2026
Undergraduate Computer Engineering
- GPA: 3.88/4.00
- Achievements:
  - Best Presentation Team - Bangkit Academy 2024 Batch 2 (National Scale)
  - Third Runner-Up for Paper Competition - AEROSPACE 2024 (National Scale)
  - Second Runner Up for Best Essay Competition - UNLIMITED 2023 (National Scale)

Publication

- **GA-SAR-Det: A Multi-scale YOLOv11 Architecture for Small and Densely Distributed SAR Ship Target Detection (2026).** Published in International Journal of Intelligent Engineering and Systems. DOI: 10.22266/ijies2026.0930.71

Projects

- **YOLOv5 Backbone Comparison on Plant Detection (2026)**: Reconstructed YOLOv5 with four CNN backbones (EfficientNetB1, ResNet152, VGG16-BN, MobileNet-Large) to identify plant species from field-collected leaf images.
- **SAR Ship Detection Research** (2025): Built and evaluated a lightweight deep learning model by reconstructing YOLO with C3Ghost module for small-target detection in maritime SAR imagery.
- **IoT-Based Essential Oil Distillation System** (2025): Developed the electronic control system and IoT integration for a water-cooling mechanism in an essential oil distillation device, enabling automated temperature regulation and real-time monitoring.
- **Harmful Content Filtering Extension** (2025): Created a Chrome extension that filters toxic comments on Twitter using a fine-tuned IndoBERT model with Hugging Face Inference API integration.
- **AGV Medical Assistant** (2025): Designed a 3D-CNN for human recognition, implemented the A* pathfinding algorithm, and developed a position-tracking database using an Indoor Positioning System (IPS) for an autonomous guided vehicle (AGV) robotics system.This project aimed to reduce the workload of health workers.
- **SiBantu** (2024): A location-based service platform utilizing machine learning for matching users with home repair service providers.
- **Comparison Machine Learning Methods in Plant Recommendation System** (2024): Demonstrated that supervised learning approaches are most suitable for enhancing recommendation accuracy. Presented findings and methodology at AEROSPACE 2023.
- **DF Seperator** (2023): A conceptual waste separator system featuring Waste Image Recognition powered by a Data Fusion algorithm. This project showcased proficiency in computer vision and aimed to improve waste sorting efficiency.
- **Plant Recommendation** (2023): An integrated multi-sensor data and machine learning to provide tailored plant cultivation advice.
- **Recreation Recommendation Program** (2023): Develop a Recreation Recommendation system to help tourists exploring local landmarks using Python.

Organizational Experience

TEDxUNDIP 4.0 Oct 2024 - May 2025
Project Manager
- Directed TED Talks main event, orchestrating all speaker sessions, entertainment segments (traditional dance, theater), and event rundown.
- Formulated the concept and managed the exhibition featuring 5 sensory-cultural themed booths.

Certificates

- **Advanced Learning Algorithm (2024)**: Deeper exploration of optimization and ML algorithms.
- **Convolutional Neural Network in TensorFlow (2024)**: Building and training CNNs for computer vision.
- **Custom and Distributed Training with TensorFlow (2024)**: Scaling ML models with TensorFlow strategies.
- **Advanced Computer Vision with TensorFlow (2024)**: High-level vision tasks and architectures.
- **Build Basic Generative Adversarial Networks (GANs) (2024)**: Introductory GANs for image generation.
- **Custom Models, Layers, and Loss Functions with TensorFlow (2024)**: This course covers custom models, layers, and loss functions using TensorFlow.
- **Data Pipelines with TensorFlow Data Services (2024)**: This course focuses on building data pipelines with TensorFlow Data Services.

Language Proficiency Test

- **TOEFL** (2025): Listening Comprehension: 63, Structure and Written Expression: 52, Reading Comprehension: 67, Total Scores: 607
- **IELTS** (2022): Listening: 7.5, Reading: 8.0, Writing: 6.0, Speaking: 6.0, Overall Band Score: 7.